Answer Key - Chemistry - Graduate Level Assessment 24 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** E
   Options: A) AlCl 3(aq); B) CBr 4(aq); C) NaOH(aq); D) HCl(aq); E) LiOH(aq)
   Note: Solid MgCl 2 is most soluble in LiOH(aq) because the presence of hydroxide ions from LiOH promotes the formation of soluble complexes with magnesium ions, enhancing the dissociation of MgCl 2 in solution.

2. **Answer:** A
   Options: A) 4.26; B) 3.66; C) 9.26; D) 2.66; E) 10.52
   Note: The pKa of a weak acid is equal to the pH of a solution when the concentration of the acid equals its dissociation constant, and in this case, the pH of 4.26 indicates that the concentration of the dissociated hydrogen ions is equal to the concentration of the undissociated acid, thus making the pKa also 4.26.

3. **Answer:** A
   Options: A) 400 $\mathrm{N \cdot m^{-1}}$; B) 350 $\mathrm{N \cdot m^{-1}}$; C) 250 $\mathrm{N \cdot m^{-1}}$; D) 285 $\mathrm{N \cdot m^{-1}}$; E) 450 $\mathrm{N \cdot m^{-1}}$
   Note: The correct answer of 400 N · m¹ is derived from the relationship between the vibrational frequency of the molecular bond, as given by the infrared line at 2309 cm¹, and the force constant, which can be calculated using the formula $k = 4\pi^2\mu\nu^2$, where μ is the reduced mass of the hydrogen iodide molecule and ν is the wavenumber corresponding to the observed infrared transition.

4. **Answer:** A
   Options: A) 1.75; B) 3.50; C) 14.00; D) 10.00; E) 12.00
   Note: The reduced mass of a nitrogen molecule, composed of two nitrogen atoms each with an atomic mass of 14.00, is calculated using the formula $\mu = \frac{m_1 m_2}{m_1+m_2}$, which results in $\mu = \frac{14.00 \cdot 14.00}{14.00+14.00} = 7.00$ u, and when expressed in terms of kg, it is approximately 1.75 kg for the system's dynamics.

5. **Answer:** A
   Options: A) 3.0×10^5 Joules, 3.0×10^12 ergs; B) 2.5×10^5 Joules, 2.5×10^12 ergs; C) 2.0×10^5 Joules, 2.0×10^12 ergs; D) 1.8×10^5 Joules, 1.8×10^12 ergs; E) 3.56×10^5 Joules, 3.56×10^12 ergs
   Note: The correct answer is derived from the kinetic energy formula KE = 0.5 * m * v², where the weight of the automobile is converted to mass in kilograms, and the speed is converted from miles per hour to meters per second, resulting in 3.0×10^5 Joules and 3.0×10^12 ergs.

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: The statement is false because alkenes have a trigonal planar configuration around their double bond carbon atoms due to the sp^2 hybridization, which allows for 120-degree bond angles rather than the tetrahedral 109.5-degree angles associated with sp^3 hybridization.

7. **Answer:** False
   Options: A) True; B) False
   Note: The number of allowed energy levels for the 55 Mn nuclide is not limited to five due to the complex interplay of its nuclear structure and the presence of multiple nuclear states, which can significantly exceed that number.

8. **Answer:** True
   Options: A) True; B) False
   Note: The statement is true because the constants a' and b' in the Dieterici equation are indeed derived from the critical constants, where a' represents the interaction energy between molecules at the critical point and b' reflects the volume occupied by the molecules at that point, thus establishing their relationship to the critical parameters of the substance.

9. **Answer:** True
   Options: A) True; B) False
   Note: In the electrolysis of molten magnesium bromide, magnesium ions gain electrons at the cathode to form magnesium metal, while bromide ions lose electrons at the anode to produce bromine gas, confirming the statement is true.

10. **Answer:** False
    Options: A) True; B) False
    Note: The statement is false because the calculation of the number of photons in a light pulse of 2.00 mJ at a wavelength of 1.06 m yields approximately 1.13 $\times$ $10^{\{}19\}photons, not 1.80Ö10^{\{}16\}$.

Long Form Response

11. **Answer:** The orbital angular momentum quantum number, l, plays a crucial role in determining the ionization energy of an atom by influencing the spatial distribution and shielding effects of electrons. In the case of ground-state aluminum, which has an electronic configuration of [Ne]3 $s^2$3 p^1, the electron that is most easily removed during ionization corresponds to the 3 p orbital, where l equals 1. This electron is more shielded by the inner electrons and experiences less effective nuclear charge compared to the 3 s electrons, resulting in a lower ionization energy. The significance of l is thus highlighted, as it helps explain why the 3 p electron is more easily ionized, reflecting both the energy levels and the stability of electron configurations. Understanding this

relationship is vital for predicting reactivity and bonding in aluminum and similar elements.

Note: correct because the orbital angular momentum quantum number, l, influences the ionization energy by determining the electron's spatial distribution and shielding, with the 3 p electron in aluminum being the most easily removed due to its higher shielding and lower effective nuclear charge compared to the more tightly bound 3 s electrons.

12. **Answer:** At$_{4000}$ meters altitude, the atmospheric pressure decreases significantly, which directly influences the boiling point of water. The boiling point of a liquid is the temperature at which its vapor pressure equals the surrounding pressure; thus, as altitude increases and pressure decreases, the boiling point lowers. At$_{4000}$ meters, the expected boiling point of water is approximately $86.2 \cdot Thisdecreasecanbeattributedtotheloweratmosphe$
$Note:correctbecauseitaccuratelyexplainsthattheboilingpointofwaterdecreaseswithaltitudeduetored$

End of Answer Key